B.TECH CSE · MAJOR PROJECT · OCTOBER 2026

Pramana

प्रमाण *proof; a means of valid knowledge*

AI-assisted legal research for Indian criminal law that verifies every citation it makes, across the IPC → BNS transition.

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Criminal legal research in India is still done by hand.

WHO DOES IT

- Advocates, especially district-court practitioners
- Law students
- Legal-aid clinics

1
Bare act
find candidate sections

→

2
Commentary
read the ingredients

→

3
Search judgments
keyword search for precedent

→

4
Manual reading
hours per matter
[[VERIFY: time per matter]]

AND NOW

Two criminal codes since 1 July 2024
IPC and BNS in use side by side

AI tools invent citations
a wrong section is not a cosmetic error

Precedent is split across codes
new BNS sections have almost no case law yet

On 1 July 2024 India renumbered its criminal law.

INDIAN PENAL CODE, 1860 offences before 1 July 2024	SECTION	BHARATIYA NYAYA SANHITA, 2023 offences from 1 July 2024
Punishment for murder	302	Uttering words with deliberate intent to wound religious feelings
Punishment for murder by a life-convict	303	Theft
Cheating and dishonestly inducing delivery of property	420	*No such section. The BNS ends at 358.*

A section number means nothing without its code, and *the offence date decides the code.*

AI models cite law with confidence, including law that does not exist.

A LANGUAGE MODEL, ASKED DIRECTLY

Q What does BNS Section 303 cover?

A A public-panic offence: making or spreading a bomb threat.

× Wrong. BNS 303 is **Theft**.

20 *of* **68**

citations named sections not in the BNS when the model answered from memory.

15 development cases, retrieval withheld.

17–33%

of queries hallucinated by commercial legal AI research tools. [[VERIFY: Magesh et al. figure]]

PROBLEM STATEMENT

Build a research assistant for Indian criminal law that identifies applicable sections and precedents across the IPC→BNS transition, and *never presents a non-existent citation*.

What exists, and what nobody combines.

RESEARCH [[VERIFY: paper details]]

- **RAG** · Lewis et al., NeurIPS 2020
  grounds answers in retrieved text; no citation check
- **LeSICiN** · Paul et al., AAAI 2022
  graph statute identification; supervised, IPC only
- **IL-TUR** · Joshi et al., ACL 2024
  Indian legal benchmark; IPC-era labels
- **Legal AI hallucination** · Magesh et al., 2024
  measures the failure, does not prevent it
- **BGE-M3** · Chen et al., 2024
  general embedder; untested on Indian law

EXISTING TOOLS

	IPC/BNS BY DATE	VERIFIES CITATIONS	PRECEDENT ACROSS CODES	COUNTER-ARGUMENTS	
Indian Kanoon	✗	—	✗	✗	[[VERIFY]]
Manupatra / SCC Online	?	—	?	✗	[[VERIFY]]
jhana.ai	?	✓	?	?	[[VERIFY]]
NyayAssist	?	✓	?	?	[[VERIFY]]
Pramana	✓	✓	✓	✓	

— search engines and databases return real documents; they do not interpret facts.

THE GAP

No system combines **date-based code routing**, **graph-verified citations**, an **IPC→BNS precedent bridge**, **counter-arguments** and **outcome statistics**.

Facts and a date go in. A checked research memo comes out.

CASE FACTS · THE RUNNING EXAMPLE

The complainant parked his motorcycle outside his house at night. In the morning it was gone. CCTV shows the accused pushing it away down the lane and riding off. The accused had no permission to take it and did not return it.

OFFENCE DATE 15 Aug 2024

Pramana

CODE	BNS, because the offence is on or after 1 July 2024
CHARGE	BNS 303 Theft ✓ verified
PRECEDENTS	3 judgments, found through its IPC predecessor, IPC 379
DEFENCE	A counter-argument, its quote checked against the facts
OUTCOMES	How 158 appeals under IPC 379 were decided

The right code, every time
the offence date decides IPC or BNS

Every citation verified
a section is shown only if it exists in that code

Precedent across both codes
BNS charges reach IPC case law

Four steps: route, retrieve, verify, ground.



01

Route

The offence date picks the code: IPC or BNS. A fixed rule, no AI.

02

Retrieve

Find candidate sections from statute text and from similar past judgments.

03

Propose & verify

The AI proposes sections; the citation graph confirms each one exists.

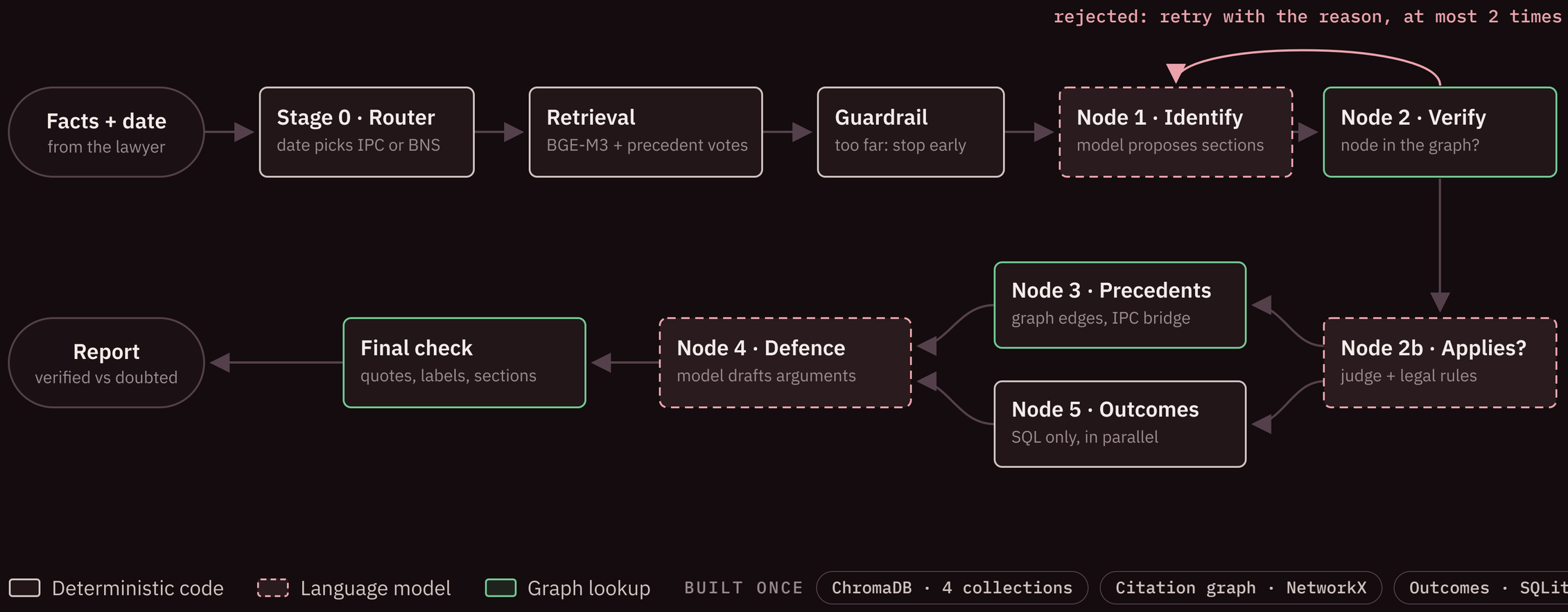
04

Ground & argue

Test each section on the facts, find precedents, draft counter-arguments.

The AI proposes. *The graph decides what is real.*

The model proposes. The graph decides.



Retrieval finds the candidates; nothing is fine-tuned.

WHY RAG, NOT FINE-TUNING

Knowledge stays current
new judgments are added, not trained in

Every answer traces to a source
each candidate is a real indexed section

The model reads the real text
the section itself is in front of it when it proposes

THE MOTORCYCLE CASE, STEP BY STEP

1. **Date rule**
   15 Aug 2024 ≥ 1 Jul 2024 → **BNS**

2. **Embed the facts**
   BGE-M3, 1024 dimensions

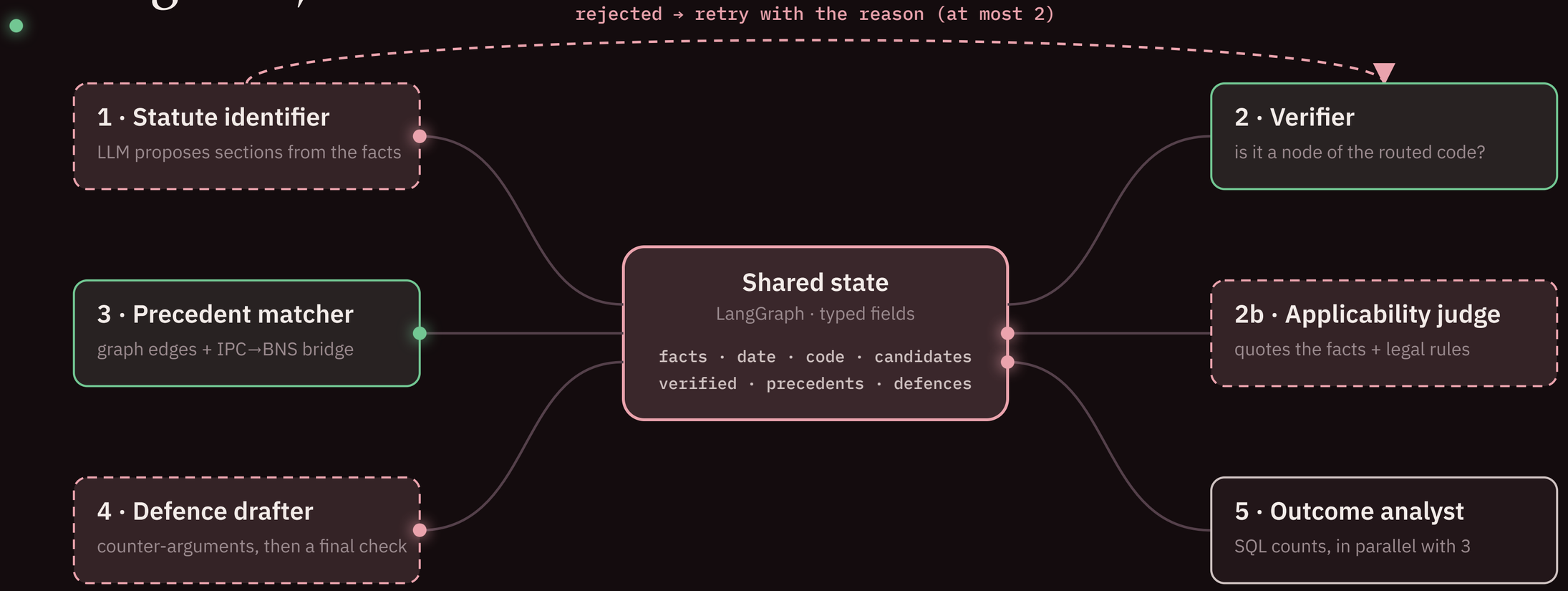
3. **Statute search + precedent votes**
   similar judgments vote for the sections they applied; fused by reciprocal rank fusion

4. **14 candidates for the agents**

303(1)	Theft	0.0320 · T3 V2
317(5)	Stolen property	0.0310 · T6 V3
324(1)	Mischief	0.0265 · T5 V30

Guardrail: closest judgment 0.686, above the 0.576 threshold, so the agents are asked.

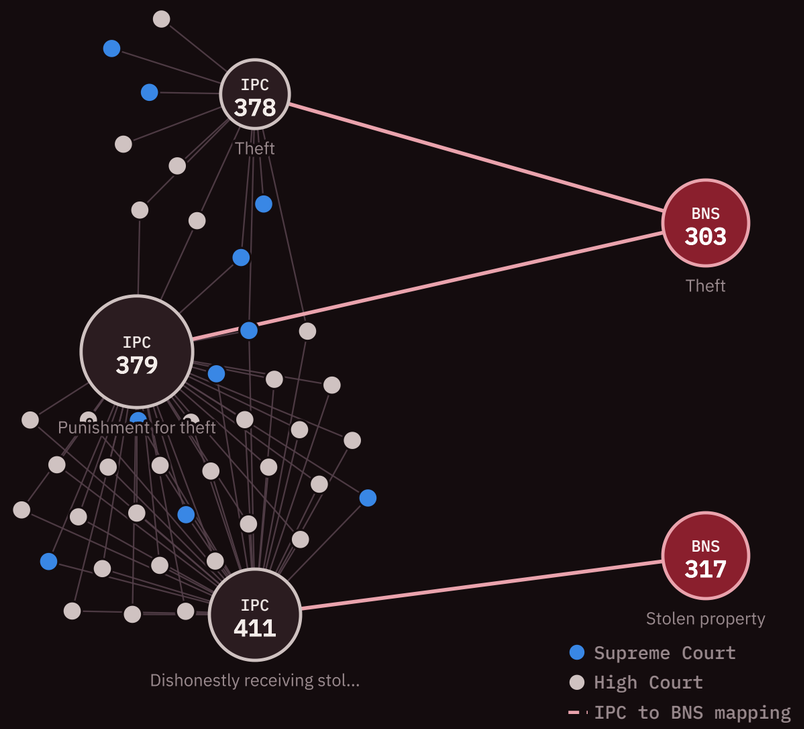
Six agents, one shared state.



Code · Language model · Graph lookup

Each agent reads and writes only its own fields of the state, so every step can be checked on its own.

Every citation is checked against the law itself.



933 sections · 2,699 judgments · 16,493 citation edges · 539 IPC→BNS bridges

THE RETRY LOOP · A REAL RUN WITH RETRIEVAL WITHHELD

ROUND 0 — identifier: 421

✕ Not a section of the BNS, which has sections 1 to 358.

ROUND 1 — identifier: 378

✕ An IPC number. Hint from the graph: its BNS counterpart is **BNS 303**.

ROUND 2 — identifier: 303

✓ **BNS 303 · Theft.** A node of the BNS. Verified.

20 / 20
real vs invented section numbers told apart

0
rejected citations ever reached a report

A real section can still be the wrong charge.

FACTS · KNIFE STABBING · 3 NOV 2024

After a quarrel over a parking space, the accused took a kitchen knife from his car and stabbed the complainant in the abdomen. Surgeons repaired a perforated intestine, and the medical report records the injury as dangerous to life.

LEGAL RULES
- death needed for homicide
- 5+ people for rioting
- lesser offence included
- weapon + injury grade

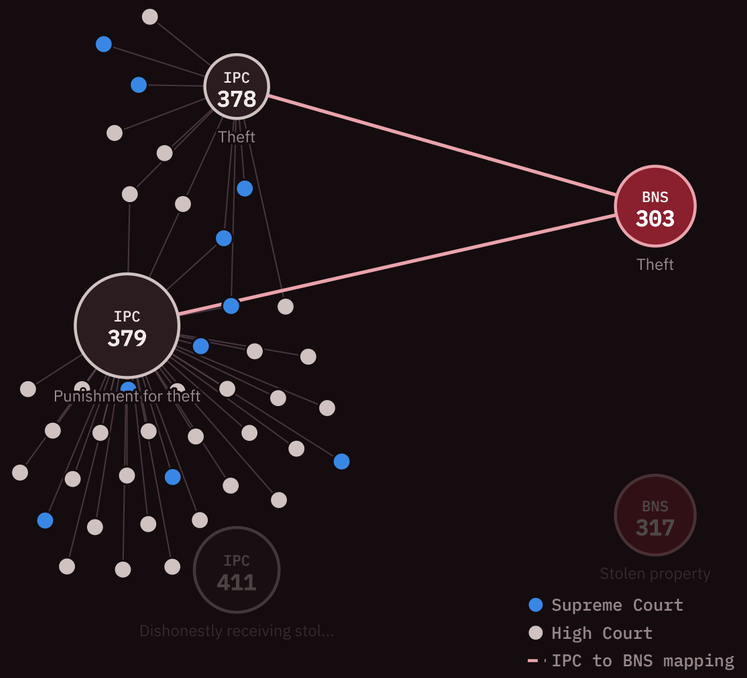
BNS 109	Attempt to murder	SUPPORTED
BNS 118(2)	Hurt by a dangerous weapon	SUPPORTED
BNS 105	Culpable homicide requires a death; the facts report grievous hurt, not death	DOUBTED
BNS 117	Grievous hurt	IN 118(2)

AGENT 4 · DEFENCE DRAFTER

3 counter-arguments ✓ each targets a charge

✓ every quote verbatim from the facts ✓ only supplied precedents cited

New law has almost no case law yet. We bridge it to the old.



BNS 303	0	judgments cite it directly
IPC 378	11	via the mapping
IPC 379	158	via the mapping

TOP 3 BY SIMILARITY TO THE FACTS

P1	Telangana High Court	0.686	Allowed
P2	Himachal Pradesh High Court	0.676	Allowed
P3	Delhi High Court	0.658	Dismissed

Agent 5 counts how appeals under each section were decided with plain SQL: IPC 379, 158 judgments: 76 allowed, 76 dismissed. Under 5 judgments, a figure is not shown.

The technology stack, and why each piece.

INTERFACE

FastAPI + web app

typed endpoints and live, stage-by-stage progress

ORCHESTRATION

LangGraph

vs one big prompt: a retry edge, parallel agents, and a shared state checked step by step

INTELLIGENCE

qwen3.5:4b via Ollama · BGE-M3

qwen3.5:4b: open-weight model with JSON-schema-constrained answers

BGE-M3 vs a legal-domain embedder: recall@3 0.533 vs 0.20, measured

KNOWLEDGE

ChromaDB · NetworkX · SQLite

ChromaDB vs FAISS / Pinecone: persistent, with metadata filters

NetworkX: exact existence check · SQLite: exact counts

Python 3.12	**PyTorch** CUDA	**sentence-transformers** embedding
BGE-M3 1024-d vectors	**ChromaDB** vector store	**NetworkX** citation graph
LangGraph agents + state	**Ollama** qwen3.5:4b	**FastAPI** API + web app
SQLite outcomes	**PyMuPDF** PDF parsing	**pytest** 422 tests

07 · DEMO

Let's run it.

THE CASE

The complainant parked his motorcycle outside his house at night. In the morning it was gone. CCTV shows the accused pushing it away down the lane and riding off. The accused had no permission to take it and did not return it.

OFFENCE DATE 15 August 2024

WATCH FOR router picks BNS · 303 ranked first · verified in the graph · precedents via IPC 379

Thank you.

Questions welcome

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▶ LIVE

Switching to the browser

http://127.0.0.1:8000